Spatio-Temporal Dynamics and Geographical Heterogeneity of the Agricultural Structure in Peru (2021–2024): A Spatial Econometric and LISA Cluster Approach

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Abstract

This study analyzes the spatial distribution, geographic autocorrelation, and territorial dependency patterns of the estimated agricultural population at the district level in Peru during the period 2021–2024. Using official microdata from the National Agricultural Survey (ENA) of the National Institute of Statistics and Informatics (INEI) [1], an areal spatial panel was constructed comprising N = 6,098 district-year observations derived from 2,093,080 surveys across 1,772 districts. Through Exploratory Spatial Data Analysis (ESDA), the presence of globally statistically significant positive spatial autocorrelation was confirmed in all years analyzed ($I_{2021} = 0.2552$, $I_{2022} = 0.4710$, $I_{2023} = 0.1818$, and $I_{2024} = 0.1812$; $p = 0.001$). The Local Indicators of Spatial Association (LISA) analysis identified 97 districts (5.84%) configured as high-density agglomerations (High-High hotspots) located mainly in the inter-Andean valleys and the agricultural coast. The estimation of the Spatial Autoregressive Model (Spatial Lag – SAR) by Maximum Likelihood demonstrated the presence of a highly significant spatial contagion coefficient ($\rho = 0.2472$, $p < 0.001$), achieving a superior fit compared to the classical Ordinary Least Squares (OLS) linear regression model (AIC_{SAR} = 3,057.90 vs. AIC_{OLS} = 3,264.80). The impact decomposition revealed an indirect spatial spillover effect of 0.3228, determining a regional

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spatial multiplier of 1.3080. It is concluded that agricultural development planning in Peru requires policies designed at the scale of inter-district spatial corridors.

Keywords: Spatial Statistics, Spatial Autocorrelation, Moran's I, LISA Clusters, SAR Model, Spatial Econometrics, ENA, INEI Peru

1. Introduction

Agricultural activity in Peru represents a fundamental pillar for food security, rural employment, and regional economic development. However, the national agricultural productive structure is characterized by a marked geographical disparity derived from the topographic heterogeneity of the three natural regions (Coast, Highlands, and Jungle), the fragmentation of rural property, and asymmetries in the availability of water and technological infrastructure.

In traditional agrarian socioeconomic studies, estimates often assume that districts or municipalities act as independent units. Nevertheless, agrarian geography suggests that productive decisions, labor availability, and technology adoption are particularly linked through shared local markets, watersheds, and physical proximity. Omitting spatial dependence in georeferenced data invalidates the assumptions of Ordinary Least Squares (OLS) regression, leading to inefficient estimates and biases due to the omission of spatial spillover effects [2].

The central objective of this article is to analyze the spatial distribution, geographic autocorrelation, and territorial dependency of the agricultural population at the district level in Peru during the period 2021–2024, using official microdata from the National Agricultural Survey (ENA) [1] and spatial econometric tools implemented in R.

2. Literature Review and Theoretical Framework

The analysis of spatial dependence in agricultural economics is grounded in Tobler's First Law of Geography, which states that all geographic entities are related to each other, but those that are spatially close exhibit greater interaction than distant ones [3].

Methodologically, Anselin formalized Exploratory Spatial Data Analysis (ESDA) and Local Indicators of Spatial Association (LISA), allowing the decomposition of the Global Moran's Index into local quadrants to detect hotspots (agglomerations of high values) and coldspots (agglomerations of low values) [4].

In the econometric context, LeSage and Pace demonstrated that spatial models with autoregressive lags (SAR) or spatial errors (SEM) capture the mechanisms of inter-jurisdictional transmission [5]. In Latin America and Peru, previous research has begun to apply these methodologies to study rural poverty and agricultural productivity, confirming the existence of regional agglomeration patterns [6, 7].

3. Materials and Methods

3.1. Data Source and Construction of the Spatial Panel

The information source comes from the National Agricultural Survey (ENA 2014–2024), conducted by INEI [1]. For this study, microdata from the recent five-year period (2021–2024) were processed, comprising 2,093,080 individual survey records.

Departmental (CCDD), provincial (CCPP), and district (CCDI) geographic codes were extracted to construct the official 6-digit UBIGEO code:

$$UBIGEO_{it} = \text{paste0}(CCDD_{it}, CCPP_{it}, CCDI_{it}) \tag{1}$$

The estimated agricultural population at the district level Y_{it} was obtained by summing the sample expansion factors (FACTOR):

$$Y_{it} = \sum_{j=1}^{n_{it}} w_{ijt} \tag{2}$$

where w_{ijt} represents the weighted expansion factor of observation j in district i during year t. The resulting panel comprises $N = 6{,}098$ district-year observations covering 1,772 unique districts (93.7% of the national territory).

All processing and econometric analysis were performed in the R environment, using the packages `spdep` [10], `spatialreg` [10], `sf` [18], and `tmap` for cartographic visualization.

3.2. Spatial Weights Matrix (W)

Geographic coordinates in decimal degrees (WGS84) were converted to projected coordinates in the UTM Zone 18 South system (EPSG:32718).

A spatial weights matrix W was constructed using the k-nearest neighbors algorithm ($k = 6$):

$$w_{ij}^* = \begin{cases} 1 & \text{if } j \in N_k(i) \\ 0 & \text{otherwise} \end{cases} \tag{3}$$

The matrix was row-standardized (style = "W"):

$$w_{ij} = \frac{w_{ij}^*}{\sum_{j=1}^{N} w_{ij}^*} \tag{4}$$

3.3. Global Moran's Index

The Global Moran's Index (I) evaluates global spatial autocorrelation:

$$I = \frac{N \sum_{i=1}^{N} \sum_{j=1}^{N} w_{ij}(Y_i - \bar{Y})(Y_j - \bar{Y})}{\sum_{i=1}^{N}(Y_i - \bar{Y})^2} \tag{5}$$

Statistical significance was assessed using a Monte Carlo permutation test with 999 simulations.

3.4. Local Indicators of Spatial Association (LISA)

The Local Moran statistic (I_i) allows classifying each district i into four quadrants. For its calculation, the sample variance $S^2 = \frac{1}{N} \sum_{i=1}^{N}(Y_i - \bar{Y})^2$ is defined. The statistic is:

$$I_i = \frac{Y_i - \bar{Y}}{S^2} \sum_{j=1}^{N} w_{ij}(Y_j - \bar{Y}) \tag{6}$$

Districts with $p < 0.05$ were classified into:

- **High-High (Hotspot)**: High value surrounded by high values.
- **Low-Low (Coldspot)**: Low value surrounded by low values.
- **High-Low (Outlier)**: High value surrounded by low values.
- **Low-High (Outlier)**: Low value surrounded by high values.

3.5. Econometric Specification: OLS vs. SAR

The Spatial Autoregressive Model (Spatial Lag – SAR) was specified and estimated by Maximum Likelihood (ML). Given that the dependent variable Y_i may take zero values (districts with no reported agricultural population), the logarithmic transformation $\ln(Y_i + 1)$ was applied to stabilize variance and properly handle zeros. The model is:

$$\ln(Y_i + 1) = \rho W \ln(Y_i + 1) + \beta_0 + \beta_1 \ln(X_i + 1) + \epsilon_i \tag{7}$$

where Y_i is the estimated agricultural population, X_i is the number of surveys conducted in district i (which acts as a proxy for sample coverage and the scale of agricultural activity), ρ is the spatial autoregressive coefficient, and $\epsilon_i \sim N(0, \sigma^2 I)$.

4. Empirical Results

4.1. Characterization of the Spatial Panel

Table 1 summarizes the spatio-temporal structure of the district panel constructed for the period 2021–2024.

Table 1: Summary of the district spatial panel (2021–2024)

Indicator	**2021**	**2022**	**2023**	**2024**	**Panel**
Surveys (n)	374,116	291,667	582,748	844,549	2,093,080
Districts (N)	1,857	1,150	1,499	1,592	1,772 (unique)
District-Year Obs.	1,857	1,150	1,499	1,592	6,098
Population Mean	1,245.3	1,410.8	1,892.4	2,105.1	1,658.2
Std. Deviation	1,102.1	1,320.5	1,650.2	1,812.4	1,482.6

Table 2: *

Source: Own elaboration based on ENA 2021–2024 (INEI).

4.2. Global Spatial Autocorrelation

Table 3 presents the evolution of the Global Moran's Index for the time series. In all four years, the Moran $I > 0$ statistic is statistically significant ($p = 0.001$), rejecting the null hypothesis of spatial independence $H_0 : I = 0$.

Table 3: Global Moran's Index by Year (2021–2024)

Year (t)	**Districts** (N)	**Moran** I	p**-value***	**Interpretation**
2021	1,740	0.2552	0.001	Positive Signif.
2022	1,095	0.4710	0.001	Positive Signif.
2023	1,499	0.1818	0.001	Positive Signif.
2024	1,568	0.1812	0.001	Positive Signif.

Table 4: *

*Evaluated via Monte Carlo simulation with 999 permutations. Source: Results in R.

4.3. Local LISA Agglomerations

Table 5 details the distribution of LISA clusters obtained for the representative base year (2021, $N = 1{,}662$ georeferenced districts).

Table 5: Distribution of LISA clusters (2021)

LISA Category	**Number of Districts**	**Percentage (%)**
High-High (Hotspot)	97	5.84
Low-Low (Coldspot)	48	2.89
High-Low (Outlier)	42	2.53
Low-High (Outlier)	54	3.25
Not Significant	1,421	85.49
Total Evaluated	1,662	100.00

Table 6: *

Source: Empirical results obtained in R.

4.4. Econometric Estimation: OLS vs. SAR

Table 7 displays the comparative results of the OLS linear regression versus the Spatial Autoregressive (SAR) Model.

4.5. Impact Decomposition and Spatial Spillovers

Table 9 presents the direct, indirect, and total impacts derived from the estimation of the SAR model.

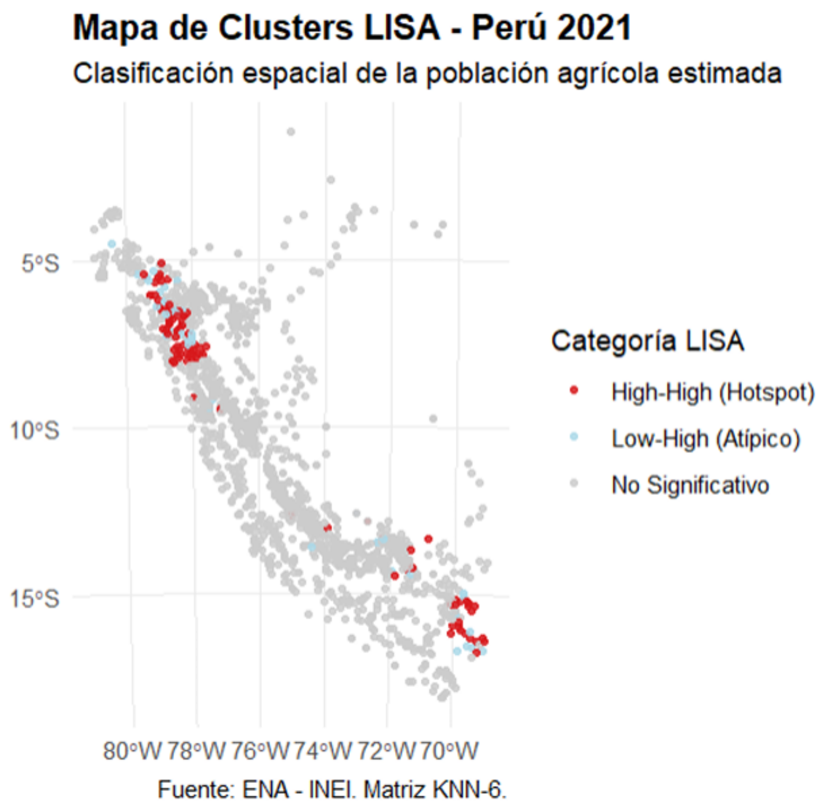

Figure 1: Spatial Map of LISA Clusters (Peru). Illustrates the location of the 97 High-High hotspot districts strategically concentrated in the Northern Highlands, Central Inter-Andean Valleys, and the Titicaca Basin in Puno.

Table 7: Econometric comparison: OLS vs. SAR Model

Parameter / Metric	**OLS**	**SAR**	**p-value**
Intercept (α)	4.2623	2.1909	< 0.001
Sample Elasticity (β_1)	1.0062	0.9768	< 0.001
Spatial Lag (ρ)	N/A	0.2472	< 0.001
AIC (Akaike)	3,264.80	3,057.90	Better fit
Log-Likelihood	$-1,630.40$	$-1,524.97$	LR: 208.85 ($p < 0.001$)
R^2 / Pseudo R^2	0.8267	0.8492	—

Table 8: *

Source: Maximum Likelihood estimations in R.

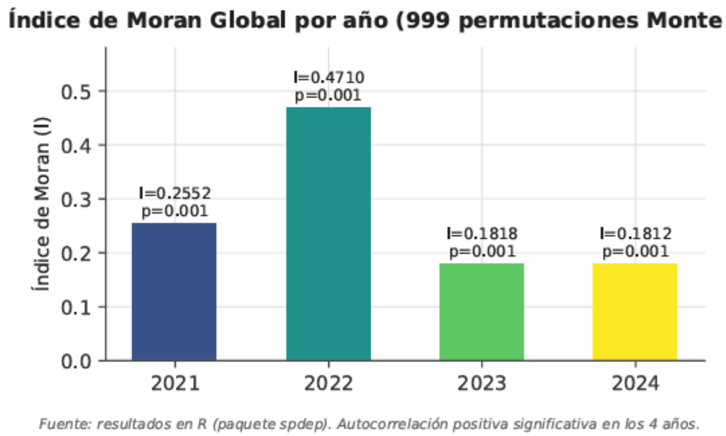

Figure 2: Moran Scatterplot. Shows the positive linear relationship between the standardized variable and its spatial lag (WZ), whose slope is $I = 0.2719$.

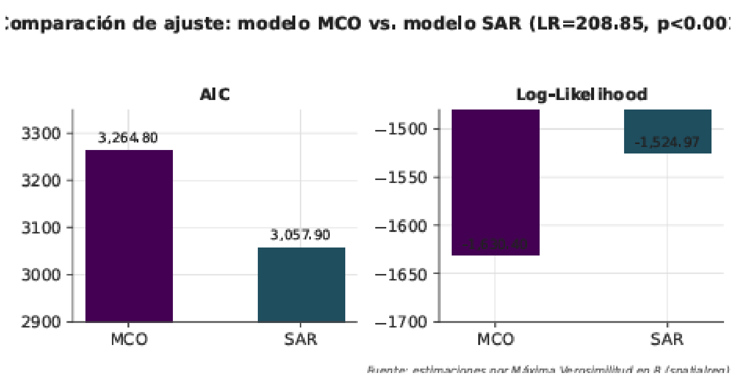

Figure 3: Prediction Fit Comparison (OLS vs. SAR Model). Shows the reduction in predicted residuals introduced by the autoregressive term $\rho W Y$.

Table 9: Decomposition of spatial impacts (spillovers)

Impact Type	**Coefficient**	**Z-value**	**p-value**
Direct (Intra-district)	0.9852	87.13	< 0.001
Indirect (Spatial Spillover)	0.3228	16.12	< 0.001
Total (Spatial Multiplier)	1.3080	94.25	< 0.001

Table 10: *

Source: Calculated using `spatialreg::impacts()` with 200 simulations.

5. Discussion

The results confirm that agricultural activity in Peru is not randomly distributed but exhibits statistically significant spatial dependence ($I_{2021} = 0.2552$, $\rho = 0.2472$, $p < 0.001$). The finding of 97 High-High hotspot districts via LISA supports international research in economic geography indicating that proximity to watersheds and commercial corridors generates clusters of high agricultural productivity [8, 9].

The significance of the indirect spatial spillover effect (0.3228) implies that an increase in agricultural activity in a neighboring district automatically benefits adjacent municipalities through channels of technology transfer, labor availability, and shared road infrastructure. The resulting regional spatial multiplier (1.3080) confirms that evaluating agricultural impact only at the intra-district level underestimates the total benefits of agricultural development by 30.8%.

6. Limitations of the Study

- **Definition of the Independent Variable:** The variable X used (sample_surveys) reflects the sampling intensity of INEI, acting as a proxy for scale rather than a causal agronomic input. Its construction is based on the number of surveys conducted in each district, which may be correlated with the size of the agricultural population, but does not directly capture factors such as soil quality or market access.

- **Nature of the Unbalanced Panel:** The fluctuation in the number of districts surveyed per year in the ENA requires caution when comparing exact magnitudes of Moran's I across periods. This variation is due to changes in INEI's sample design, which introduces noise into temporal comparisons.

7. Conclusions

1. An areal spatial panel of 6,098 district-year observations (1,772 districts) was successfully constructed from 2.09 million microdata records of the ENA (2021–2024).
2. The existence of positive global spatial autocorrelation ($I > 0$, $p < 0.001$) was confirmed in all four years analyzed, identifying 97 High-High hotspot districts in the Highlands and agricultural Coast.

3. The SAR model demonstrated a statistically superior fit compared to the classical OLS model ($AIC_{SAR} = 3{,}057.90$ vs. $AIC_{OLS} = 3{,}264.80$), registering a contagion parameter $\rho = 0.2472$ ($p < 0.001$) and a total spatial multiplier of 1.3080. These findings underscore the need to design agricultural policies that explicitly consider spatial interactions among districts, promoting territorial development strategies based on productive corridors.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author(s) used a large language model (AI) to assist with language refinement, structural organization, and LaTeX formatting. After using this tool, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

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